

FLEURS test split: en_us, hi_in, bn_in, ta_in, ur_pk; ≤ 35 s per utterance, up to 600 utterances per language

Whisper-large-v3
canonical; float16, beam 5

Indic fine-tunes (large-v2)
Hindi, Bengali, Tamil

Two text normalisers: *Ours* (NFC, lowercase, punctuation strip); *Whisper-Radford* (English / Basic normaliser)

WER per language
95% bootstrap CI; unpaired
test vs English, BH-adjusted q

Word-level error typology
five mechanisms
(on the *Ours* normalisation)

Case study: separate track, no WER computed

**Anecdotal
Bengali clip**
98 s, consented

Whisper-large-v3
3 language hints:
none / bn / en

**Qualitative
comparison**
no reference
transcript